



Fig. 8: Cellebrite Touch, a mobile forensic kit (Image courtesy: cellebrite.com)



Fig. 9: Mobile phones bagged as evidence in the UK (Image courtesy: Wikipedia)

Digital forensics and investigation tools

Forensic scientists have a long list of tools that they use to collect and analyse evidence. While different professionals at forensics departments use highly-specialised tools, there are some basic and essential tools, which every forensic scientist must know of.

The various branches of digital forensics are described below.

Computer forensics. This is the application of digital forensic and investigation to collect and analyse evidence from a computer device so that it can be produced in the court of law. Digital artefacts can be computer machines, storage devices (hard disks, memory cards and USB pen drives) or electronic documents.

Network forensics. This branch of digital forensic science collects and analyses evidence from network-connected devices like computer Ethernet controllers, routers, switches, access points, Wi-Fi devices and the like. Under this branch, network traffic is intercepted using network packet sniffers at packet level and sniffer traffic is stored in some standard format so that it can be analysed by forensic tools later to reveal the actual

activities carried out by hackers.

Mobile forensics. Application of digital forensic science to collect and analyse evidence from handheld devices like mobiles or PDAs is called mobile forensics. Digital evidences can be in the form of mobile phones, memory cards, stored data over the mobile's internal memory like logs, events, call history, SMSes or emails.

Live forensics. In this case, forensics are performed on a connected running system.

Database forensics. Database forensics is the arm of digital forensic science where analysis is done on database logs and metadata to retrieve relevant evidence.

Digital forensics and investigating teams

A digital forensic team mines the most important information from the system or network logs using forensic tools and tactics, and produces the result in terms of binary or printed formatted information known as forensic evidences.

The team may perform the following tasks:

1. Provide forensic analysis of security breaches, which can be used for investigation, prosecution or mitigation

2. Provide assistance to legal investigation teams with e-discovery
3. Identify and authenticate specific network user activities
4. Create custom data-mining solutions with document authentication
5. Analyse secure networks
6. Record and analyse raw network activity
7. Implement a continuous network-monitoring solution
8. Formulate and fulfil incident response plans to a breach or other network event
9. Conduct email/file acquisition and analysis
10. Detect and diagnose malware and virus infestations
11. Develop custom IT security practices and implementations
12. Identify and reconstruct data access by various users
13. Identify data leakage and determine its scope

Applying digital forensics

1. Commonly used in both criminal law and private investigations
2. Evidence gathering and producing the same in court
3. Discovering or recovering deleted information from electronic media
4. Detecting cyber intrusions

The following principles must be kept in mind during a forensic investigation process:

1. Make sure evidence is not modified while it is being collected, secured and transported.
2. Digital evidence should be examined only by trained professionals.
3. During seizure, transportation and storing of digital evidence, every detail should be properly written and documented, preserved and available for review. ●

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